

**SEMESTER S7**

**OPTIMIZATION TECHNIQUES**

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEECT723</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. Enable the learner to formulate engineering minima/maxima problems as optimization problems
2. Enable the learner to deploy various constrained and unconstrained optimization algorithms to obtain the minima/maxima of engineering problems

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Engineering application of Optimization – Statement of an Optimization problem–Classification, Review of basic calculus concepts –Stationary points; Functions of single and two variables; Convexity and concavity of functions –Definition of Global and Local optima – Optimality criteria, Linear programming methods for optimum design – Standard form of linear programming (LP) problem; Canonical form of LP problem; Simplex Method, Duality, Application of LPP models in engineering | <b>9</b>             |
| <b>2</b>          | Optimization algorithms for solving unconstrained nonlinear optimization problems – Search based techniques: Direct search: Fibonacci and golden section search , Hookes and Jeeves , Gradient based method: Newton's method                                                                                                                                                                                                                                                                     | <b>9</b>             |
| <b>3</b>          | Optimization algorithms for solving constrained optimization problems– direct methods – penalty function methods, barrier method -Optimization of function of multiple variables subject to equality constraints; Lagrangian                                                                                                                                                                                                                                                                     | <b>9</b>             |

|          |                                                                                                                                                                                                                 |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | function– Inequality constrained techniques-KKT conditions-constrained steepest descent method                                                                                                                  |          |
| <b>4</b> | Modern methods of Optimization– Metaheuristic techniques: Genetic Algorithms – Simulated Annealing – Particle Swarm optimization –Ant colony optimization– : Use of Matlab/Scilab to solve optimization problem | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                         | <b>Part B</b>                                                                                                                                                                                                                                                                          | <b>Total</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

## Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                  | Bloom's Knowledge Level (KL) |
|----------------|------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Formulate an optimization problem to optimize an engineering application using the principles of basic calculus. | <b>K2</b>                    |
| <b>CO2</b>     | Apply the Simplex method to solve a linear programming problem                                                   | <b>K3</b>                    |
| <b>CO3</b>     | Solve the unconstrained optimization problems using gradient based method.                                       | <b>K3</b>                    |
| <b>CO4</b>     | Apply the various optimization techniques to solve a constrained optimization problem                            | <b>K3</b>                    |
| <b>CO5</b>     | Use metaheuristic algorithms to solve constrained and unconstrained                                              | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

## CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 2   | 2   |     |     |     |     |     |     |      |      | 2    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 2    |
| <b>CO3</b> | 3   | 2   | 3   |     |     |     |     |     |     |      |      | 2    |
| <b>CO4</b> | 3   | 2   | 3   |     |     |     |     |     |     |      |      | 2    |
| <b>CO5</b> | 3   | 2   | 3   |     |     |     |     |     |     |      |      | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                               |                      |                                  |                               |
|------------|-----------------------------------------------|----------------------|----------------------------------|-------------------------------|
| Sl. No     | Title of the Book                             | Name of the Author/s | Name of the Publisher            | Edition and Year              |
| 1          | Engineering Optimization, Theory and Practice | S.S RAO              | New Age International Publishers | 4 <sup>th</sup> Edition ,2012 |

| Reference Books |                                                             |                                  |                                      |                      |
|-----------------|-------------------------------------------------------------|----------------------------------|--------------------------------------|----------------------|
| Sl. No          | Title of the Book                                           | Name of the Author/s             | Name of the Publisher                | Edition and Year     |
| 1               | Optimization Techniques and Applications with Examples      | Xin-She Yang                     | John Wiley & Sons                    | 2018                 |
| 2               | Optimization for Engineering Design Algorithms and Examples | Deb K                            | Prentice Hall India                  | 2000                 |
| 3               | Introduction to Optimization Design                         | Arora J                          | Elsevier Academic Press, New Delhi   | 2004                 |
| 4               | Linear Programming                                          | Hardley G                        | Narosa Book Distributors Private Ltd | 2002                 |
| 5               | Genetic Algorithms and engineering optimization             | Mitsuo Gen, Runwei Cheng         | John Wiley & Sons                    | 2002                 |
| 6               | An introduction to optimization                             | Edwin KP Chong, Stanislaw, H Hak | John Wiley & Sons                    | Fourth Edition, 2013 |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                                                                                                                 |
| 1                              | NPTEL<br><a href="https://www.youtube.com/watch?v=a2QgdDk4Xjw">https://www.youtube.com/watch?v=a2QgdDk4Xjw</a>                                                                                                          |
| 2                              | NPTEL<br><a href="https://www.youtube.com/watch?v=dPQKltPBLfc">https://www.youtube.com/watch?v=dPQKltPBLfc</a>                                                                                                          |
| 3                              | NPTEL<br><a href="https://www.youtube.com/watch?v=qY-gKL7GxYk">https://www.youtube.com/watch?v=qY-gKL7GxYk</a>                                                                                                          |
| 4                              | NPTEL<br><a href="https://www.youtube.com/watch?v=Z_8MpZeMdD4">https://www.youtube.com/watch?v=Z_8MpZeMdD4</a><br><a href="https://www.youtube.com/watch?v=FKBgCpJIX48">https://www.youtube.com/watch?v=FKBgCpJIX48</a> |

# **SEMESTER 8**

**Electronics Engineering (VLSI Design and  
Technology)**